

Relativistic Quantum Autonomy Project Map

An 8-week day-by-day execution plan for a C++-first summer project integrating Schwarzschild geodesics, toy spacecraft autonomy, QAOA, quantum geometry, and a computational D-module appendix.

Prepared for Robin - Summer 2026 planning document

Generated: 2026-06-10

Core rule: build one honest chain that closes. Do not turn a two-month side project into a fake PhD thesis.

Table of Contents

1. Executive Summary and Feasibility Verdict
2. Project Definition and Research Claim
3. System Architecture
4. Reading and Tool Stack
5. Outreach Strategy and Contact Map
6. Daily Schedule - 8 Weeks
7. Weekly Deliverables and Checkpoints
8. Code Repository Blueprint
9. Report and Preprint Blueprint
10. Email Templates
11. Failure Modes and Cut Rules
12. Source and Contact Reference List

1. Executive Summary and Feasibility Verdict

This document turns the project discussion into a concrete, operational map. It is designed for a student who has strong mathematical maturity and numerical-analysis background, basic-to-comfortable Riemannian geometry and differential forms, basic quantum computing, no serious GR background yet, no control theory background yet, and strong C++ preference.

The selected project is deliberately ambitious but controlled: a toy C++ simulator for a spacecraft/test particle in Schwarzschild spacetime, with a small autonomous correction loop and a quantum-assisted planner. The project is not a real flight-autonomy system, and it must not claim quantum speedup. It should claim a serious, reproducible, mathematically honest prototype.

Project in one paragraph

Build a C++-first simulator of a toy spacecraft moving in the equatorial plane of Schwarzschild spacetime. Compare Newtonian propagation with relativistic geodesic propagation; add a small autonomous correction-burn loop; encode the discrete burn-choice planning problem as a QUBO/Ising model; test a simulated QAOA solver against brute force and greedy baselines; compute the quantum geometric tensor of the variational circuit; and add a small Macaulay2/Sage appendix treating a linearized geodesic-deviation equation as a Weyl-algebra/D-module example.

Feasibility

Target	Verdict	Reason
MVP in 4 weeks	Doable	Geodesic solver + Newtonian comparison + simple correction loop + first plots.
Strong 8-week version	Doable but demanding	QUBO/QAOA, quantum geometry, D-module appendix, polished report.
Full real rocket autonomy	Not doable	Requires aerospace engineering, control theory depth, mission data, and validation.
Quantum advantage claim	Do not claim	You can benchmark QAOA as a learning/prototyping module, not as speedup.

Guiding philosophy inherited from your Map

- Learn rigorously, write weekly, extract one toy model, and build one honest artifact.
- Separate what you derive yourself, what you quote responsibly, and what you compute/prototype.
- Every week must produce written output, not just reading.
- Never read more than two serious core sources in the same week.
- If overloaded, keep definitions, core derivations, code checkpoint, and written summary; cut extensions.

2. Project Definition and Research Claim

Working title

Quantum-Assisted Relativistic Autonomy for a Toy Spacecraft in Schwarzschild Spacetime

Research question

Can a small autonomous guidance toy model be built in which Schwarzschild geodesic propagation is corrected by a discrete burn planner, and can the planner be encoded as a QUBO that is benchmarked using classical search and a simulated QAOA circuit?

Honest final claim

By the end of the summer, the project should be able to state:

I built and analyzed a toy relativistic autonomy simulator in Schwarzschild spacetime, formulated correction-burn planning as a small QUBO, tested a quantum variational optimizer on the resulting instances, studied the quantum-geometric structure of the ansatz, and connected the linearized geodesic-deviation equation to a computational D-module example.

What this project is not

- It is not a real spacecraft guidance system.
- It is not a validated aerospace controller.
- It does not prove quantum speedup.
- It does not solve general relativity or quantum gravity.
- It does not require full Kerr spacetime, real mission data, or advanced D-module theory.

Main deliverables

- GitHub repository with C++ simulation modules, tests, and reproducible experiments.
- 20-30 page main report explaining the mathematics, algorithms, experiments, and limitations.
- 6-8 page preprint-style note summarizing the project in a supervisor-facing format.
- Supervisor email package: one-page summary, reading record, current figures, and precise questions.
- A minimal slide outline can be added later if needed, but it is not part of this PDF.

3. System Architecture

Layer 1 - Relativistic dynamics

Implement the equatorial Schwarzschild test-particle model. Use geometric units initially, $G = c = 1$. Propagate either the geodesic equations directly or use a first-order conserved-quantity formulation. The first validation is not autonomy; it is conservation.

- State variables: r , ϕ , $dr/d\lambda$, $d\phi/d\lambda$, t , τ , plus diagnostic energy and angular momentum.
- Baseline comparison: Newtonian two-body propagation versus Schwarzschild propagation.
- Validation plots: orbit, $r(t)$, $\phi(t)$, energy drift, angular momentum drift, proper time versus coordinate time, periapsis precession.
- Minimum result: the relativistic model should show a small but visible drift/precession relative to the Newtonian model in chosen units.

Layer 2 - Autonomy loop

Define a toy onboard loop: observe state $\rightarrow$ estimate state $\rightarrow$ choose burn $\rightarrow$ propagate. Keep this discrete, simple, and fully synthetic.

```
state x = (r, phi, vr, vphi, clock_drift)
actions A = {no burn, +radial, -radial, +tangential, -tangential}
cost J = alpha*target_error + beta*fuel + gamma*clock_drift + delta*safety_penalty
```

- Week 3 uses a greedy planner and simple feedback correction.
- Week 4 converts a finite-horizon burn-choice problem into QUBO/Ising form.
- The autonomy claim is: toy guidance/navigation/control loop, not flight-ready autonomy.

Layer 3 - Quantum optimization

Encode a short planning horizon as binary variables $z_{\{i,k\}}$, where $z_{\{i,k\}}=1$ means action k is chosen at step i . Use penalties so exactly one action is chosen per time step. Solve small instances by brute force first; only then apply QAOA.

- Ground truth: brute force over all action sequences for tiny horizons.
- Classical baseline: greedy or local search.
- Quantum module: simulated QAOA circuit using Qulacs C++ if possible, with Python only for plotting if needed.
- Success metric: reproducible comparison, not speedup.

Layer 4 - Quantum geometry

For a parameterized circuit $|\psi(\theta)\rangle$, compute the quantum geometric tensor numerically. Its real part gives a metric on the variational parameter space. Use it to compare ordinary parameter search with a natural-gradient style update if time permits.

```
Q_ij = <partial_i psi | partial_j psi> - <partial_i psi | psi><psi | partial_j psi>
g_ij = Re(Q_ij)
theta_next = theta - eta * inverse(g + eps I) * grad(J)
```

This is the clean bridge between quantum computing and differential geometry: the state space is not just a vector space; the variational family carries geometry.

Layer 5 - D-module/Weyl-algebra appendix

Do not force D-modules onto the full nonlinear geodesic system. Use them on a linearized equation. The clean object is a scalar or low-dimensional linear ODE obtained from geodesic deviation or radial perturbations around a reference orbit.

- Derive a simplified linear differential equation for a perturbation ξ .
- Clear denominators to get polynomial coefficients.
- Work in a Weyl algebra $D = \mathbb{C}[r]\langle d_r \rangle$.
- Create the left ideal generated by the differential operator.
- Use Macaulay2 or Singular/Sage to compute a small Groebner basis/characteristic object if feasible.
- Report it as a computational appendix, not the core project.

4. Reading and Tool Stack

Reading stack - strict version

Area	Source	Extract only this
General relativity	Schutz or Carroll;ourgoulhon lecture notes	Metric, geodesics, Christoffel symbols, Schwarzschild, conserved quantities, geodesic deviation.
Numerical methods	Your advanced numerical analysis notes	ODE integration, stability diagnostics, curvature as numerical error, convergence discipline.
Quantum computing	Nielsen & Chuang plus QAOA paper	Gates, measurement, cost Hamiltonians, QAOA circuit structure.
Quantum geometry	Short notes/papers on quantum geometric tensor	Fubini-Study metric, parameterized circuits, natural gradient.
D-modules	Coutinho first; Schapira/Sabbah selectively	Weyl algebra, differential equations as modules, first examples only.
Autonomy/GNC	NASA GNC overview; ONERA DTIS context	Guidance, navigation, control vocabulary and toy-loop framing.

Tool stack

- Main language: C++20.
- Linear algebra: Eigen.
- ODE integration: your own RK4/RK45 first; Boost.Odeint optional.
- Quantum circuit simulation: Qulacs C++ first if installation is reasonable; otherwise a small self-written state-vector simulator for ≤ 8 qubits is acceptable.
- Plotting: minimal Python/Matplotlib scripts are allowed even if the core simulator is C++.
- Symbolic geometry checks: SageMath/SageManifolds optional.
- D-module computation: Macaulay2/Singular/Sage, not C++.
- Report writing: LaTeX or Markdown $\rightarrow$ PDF; repository README in Markdown.

Learning rule for tools

Do not spend a week installing tools. The hierarchy is: compile C++ simulator first; plot results second; add quantum circuit library third; add D-module tooling fourth. If Qulacs blocks you, implement a small state-vector simulator for 2-6 qubits and move on.

5. Outreach Strategy and Contact Map

The outreach plan is staged. Do not email senior researchers on Day 1 with only an idea. First produce a small artifact: a figure, a repo skeleton, or a one-page summary. Then ask narrow questions.

Contact timing

When	Action	Exact purpose
Week 1 Day 6	Draft only	Write one-page project summary. Do not send yet.
Week 2 Day 6	GR/numerical relativity first wave	Send 2 short emails: one GR/numerical-relativity contact, one relativistic timing/reference-frame contact.
Week 3 Day 5	GNC/autonomy context	If the autonomy loop works, contact one GNC-oriented person/team or ask for a PhD/postdoc referral.
Week 4 Day 5	Quantum information/QAOA first wave	Send 2 emails to QI PhD/postdocs or young researchers with QUBO/QAOA framing.
Week 5 Day 7	Follow-up window	Send polite follow-up only if no reply after 7-10 days.
Week 6 Day 5	D-module first wave	After deriving one linear ODE and writing the Weyl-algebra object, send D-module/algebraic-analysis inquiry.
Week 7 Day 5	Progress update	Send a short update with 1 figure to anyone who replied.
Week 8 Day 4	Supervisor package	Send polished package to 3-4 targeted senior people. Ask for feedback or referral, not supervision immediately.

Target categories and names to check before emailing

Important freshness note: the names below are based on public pages checked during the creation of this document. Before sending, open the source page and verify that the person is still in the relevant team and that the topic fit still makes sense.

Area	Possible contacts/teams	What to ask
Quantum information - Sorbonne/LIP6	LIP6 QI team; Damian Markham; Jessica Bavaresco; Marco Tulio Quintino; Eleni Diamanti; Elham Kashefi; Frederic Grosshans; Alex Bredariol Grilo; current PhD/postdocs listed on team/personal pages	QAOA framing, parameterized circuits, quantum verification/learning, quantum metrology if relevant.
Quantum information - Paris-Saclay/Inria/Polytechnique	Marc-Olivier Renou; PhIQuS team; Titouan Carette; Filippo Vicentini	Quantum information, quantum simulation, variational algorithms, computational quantum methods.
GR/numerical relativity	Ericourgoulhon; LUX/Observatoire de Paris; SageManifolds ecosystem	Schwarzschild geodesic modeling, numerical relativity sanity check, symbolic tensor checking.
Relativistic timing/reference frames	LTE/SYRTE; Pacome Delva; Aurelien Hees; Peter Wolf; relevant PhD/postdoc pages if available	Clock drift, relativistic navigation, time/frequency transfer, reference frames.
Aerospace GNC/autonomy	ONERA DTIS/IGNC unit; NASA GNC resources for vocabulary	Guidance-navigation-control vocabulary, toy autonomy framing, estimation/control realism.
D-modules/algebraic analysis	Pierre Schapira; Claude Sabbah; IMJ-PRG algebraic analysis; CMLS/Polytechnique	Weyl algebra appendix, linearized differential equation as module, avoid deep overclaiming.

Outreach rules

- Email one person at a time inside a team. Do not spam a whole group.
- Lead with artifact, not ambition: one figure, one precise question, one GitHub link if clean.
- Ask for a 15-minute sanity check or a referral to a PhD student/postdoc.
- Never say the project proves quantum advantage.
- Never ask for supervision in the first message unless you already have a polished package.
- If someone replies, answer within 24-48 hours with a clear thank-you and one follow-up question.

6. Daily Schedule - 8 Weeks

Assumption: 15-20 hours per week. Weekdays are 2-3 focused hours. Saturday is the deep-work block. Sunday is review/rest and writing. Shift days if needed; keep the sequence.

Week 1 - Minimal GR, repository setup, and first geodesic engine

Goal: End the week with a compiling C++ project, an RK4/RK45 integrator, and a first Schwarzschild geodesic RHS even if not yet perfect.

Day 1 - Orientation and repo skeleton

- Work block: Read the executive sections of this map. Create GitHub repo. Make folders `cpp/`, `include/`, `experiments/`, `report/`, `symbolic/`. Install compiler, CMake, Eigen. Write README with the honest project claim.
- Output: README v0; repo compiles hello-world; issue tracker with 8 weekly milestones.
- Contact action: No emails. Create a private list of 10 possible helpers.

Day 2 - Minimal GR definitions

- Work block: Read only: metric tensor, line element, proper time, geodesic equation, Christoffel symbols. Write a one-page notation sheet: coordinates, units, conventions.
- Output: GR notation sheet v1.
- Contact action: No emails.

Day 3 - ODE integration module

- Work block: Implement Vec, State, RHS interface, RK4 stepper, and a test ODE such as harmonic oscillator.
- Output: RK4 test plot/data; unit test showing bounded oscillator energy error.
- Contact action: No emails.

Day 4 - Schwarzschild model derivation

- Work block: Write equatorial Schwarzschild metric in geometric units. Derive or copy responsibly the first-order equations you will actually integrate.
- Output: Derivation note: which equations are derived, which are quoted.
- Contact action: No emails.

Day 5 - First geodesic RHS

- Work block: Implement Schwarzschild RHS. Add parameters M , initial r , ϕ , radial/tangential velocities. Save trajectory CSV.
- Output: First orbit CSV, even if ugly.
- Contact action: Draft one paragraph explaining the project for future emails; do not send.

Day 6 - Diagnostics and first figure

- Work block: Add energy/angular momentum diagnostics if using conserved quantities. Plot trajectory using a quick Python script if necessary.
- Output: Figure 1: first trajectory; diagnostic text file.
- Contact action: Draft GR email but do not send yet.

Day 7 - Checkpoint and cleanup

- Work block: Write a one-page weekly checkpoint: definitions known, derivation reproducible, what is vague, next week cuts. Clean repo.
- Output: Week 1 checkpoint PDF/MD.
- Contact action: No emails. Select 2 GR contacts for Week 2.

Week 2 - Reliable Schwarzschild simulator and Newtonian comparison

Goal: End the week with validated relativistic propagation, Newtonian baseline, and first outreach to GR/relevant timing people.

Day 8 - Validation orbit

- Work block: Choose simple initial conditions. Test circular-ish and eccentric-ish orbits. Track numerical stability versus time step.
- Output: Table: dt versus conservation drift.
- Contact action: No emails.

Day 9 - Newtonian baseline

- Work block: Implement Newtonian two-body propagator in same coordinate convention. Export comparable trajectories.
- Output: Overlay plot: Newtonian vs Schwarzschild.
- Contact action: No emails.

Day 10 - Proper time / clock drift

- Work block: Add proper-time accumulation along trajectory or a simplified clock drift diagnostic.
- Output: Plot: coordinate time versus proper time difference.
- Contact action: No emails.

Day 11 - Periapsis / drift metric

- Work block: Define a numerical drift metric: final target error, radial phase drift, periapsis shift, or clock drift.
- Output: Metric file and plot.
- Contact action: No emails.

Day 12 - Write mini technical note

- Work block: Write 2-3 pages: equations, assumptions, numerical method, validation plots.
- Output: Technical note v1.
- Contact action: Prepare two short emails with one figure attached or linked.

Day 13 - First outreach wave: GR/timing

- Work block: Send 1 email to a GR/numerical relativity person/team and 1 email to a relativistic time/reference-frame person/team. Ask for sanity check or referral.
- Output: Emails sent; outreach log created.
- Contact action: Send only if you have Figure 1 and a clean one-paragraph scope.

Day 14 - Review and issue list

- Work block: Light review. Fix documentation. Create issues for Week 3 autonomy.
- Output: Week 2 checkpoint.
- Contact action: No follow-up yet.

Week 3 - Toy autonomy loop: observe, estimate, correct

Goal: End the week with a simple correction-burn loop that reduces target error in the toy simulator.

Day 15 - Define autonomy state and target

- Work block: Choose mission toy: maintain orbit radius, reach target angle, or minimize final error after perturbation. Define state x and target x^* .
- Output: One-page autonomy specification.
- Contact action: No emails.

Day 16 - Sensor noise and perturbations

- Work block: Add synthetic measurement noise and occasional perturbation impulses. Keep distributions simple.
- Output: CSV with true state vs observed state.
- Contact action: No emails.

Day 17 - Simple estimator

- Work block: Implement moving average or simple Kalman-like correction only if easy. Otherwise use noisy state directly.
- Output: Plot: true vs estimated r or ϕ .
- Contact action: No emails.

Day 18 - Discrete burn actions

- Work block: Implement actions: no burn, +/- radial, +/- tangential. Apply small velocity changes at planning times.
- Output: Simulation can apply action sequence.
- Contact action: No emails.

Day 19 - Greedy controller

- Work block: At each decision time, try all single-step actions and choose action minimizing short-horizon cost.
- Output: Plot: uncontrolled vs greedy-controlled trajectory.
- Contact action: No emails.

Day 20 - Autonomy write-up and GNC outreach

- Work block: Write 1-2 pages explaining the toy GNC loop and limitations.
- Output: Autonomy note v1; Figure 2.
- Contact action: Optional: email one GNC/autonomy person/team or ask ONERA/IGNC-oriented contact for vocabulary/referral.

Day 21 - Checkpoint and response handling

- Work block: Review replies from Week 2. If someone answered, prepare a precise follow-up question.
- Output: Week 3 checkpoint.
- Contact action: If no reply from Week 2, do not follow up yet unless 7+ days passed.

Week 4 - QUBO formulation and first quantum optimization module

Goal: End the week with a finite-horizon burn-choice problem expressed as QUBO/Ising and solved by brute force.

Day 22 - Planning horizon definition

- Work block: Choose small horizon $H = 3$ to 6 and action count $K = 3$ to 5 . Define binary variables $z_{i,k}$.
- Output: QUBO variable map.
- Contact action: No emails.

Day 23 - Penalty design

- Work block: Add exactly-one-action penalties and cost terms for final error, fuel, safety, and clock drift.
- Output: Mathematical QUBO derivation v1.
- Contact action: No emails.

Day 24 - Brute force solver

- Work block: Implement brute force enumeration for tiny instances. This is the ground truth.
- Output: Table: best action sequence and cost.
- Contact action: No emails.

Day 25 - Classical baseline

- Work block: Implement greedy or local-search baseline and compare against brute force.
- Output: Baseline comparison table.
- Contact action: No emails.

Day 26 - Quantum circuit setup

- Work block: Install/test Qulacs C++ or implement a tiny state-vector simulator. Create a two-qubit or three-qubit test circuit.
- Output: Quantum hello-world result.
- Contact action: Prepare QI outreach email with QUBO + baseline figure.

Day 27 - First QI outreach wave

- Work block: Write the QAOA plan: cost Hamiltonian, mixer, parameters beta/gamma.
- Output: QAOA formulation note v1.
- Contact action: Send 2 emails to QI PhD/postdocs/young researchers. Ask whether the QUBO/QAOA framing is honest.

Day 28 - Checkpoint

- Work block: Review code and report. Record unresolved QAOA issues.
- Output: Week 4 checkpoint.
- Contact action: No follow-up yet.

Week 5 - QAOA experiments and benchmark discipline

Goal: End the week with QAOA running on tiny burn-planning instances and compared against brute force and greedy baselines.

Day 29 - Cost Hamiltonian implementation

- Work block: Map QUBO coefficients to Pauli-Z terms. Verify energies against brute force bitstrings.
- Output: Unit test: Hamiltonian energy equals QUBO cost.
- Contact action: No emails.

Day 30 - Mixer and circuit

- Work block: Implement $p=1$ QAOA circuit first. Use small qubit count.
- Output: Circuit output distribution for one instance.
- Contact action: No emails.

Day 31 - Parameter search

- Work block: Implement grid search over beta/gamma before fancy optimization.
- Output: Heatmap or table of energy by parameters.
- Contact action: No emails.

Day 32 - $p=2$ or repeated runs

- Work block: If $p=1$ works, try $p=2$. Otherwise improve $p=1$ plots and tests.
- Output: QAOA benchmark v1.
- Contact action: No emails.

Day 33 - Compare and write honestly

- Work block: Write 2 pages: what QAOA did, what it did not do, why no speedup is claimed.
- Output: Quantum optimization section draft.
- Contact action: If Week 2 emails have no reply, send one polite GR follow-up.

Day 34 - QI follow-up window

- Work block: Prepare a clean 5-line update to anyone who replied.
- Output: Updated QI question list.
- Contact action: If no QI reply after 7-10 days, send one polite follow-up or contact a different PhD/postdoc.

Day 35 - Checkpoint and cut decision

- Work block: Decide if QAOA is stable enough. If not, freeze $p=1$ and move to quantum geometry.
- Output: Week 5 checkpoint.
- Contact action: No new emails.

Week 6 - Quantum geometry and computational D-module appendix

Goal: End the week with a quantum geometric tensor computation and one Weyl-algebra/D-module toy object tied to linearized geodesic deviation.

Day 36 - Quantum geometric tensor derivation

- Work block: Write the formula for Q_{ij} and g_{ij} . Decide finite-difference or parameter-shift approximation.
- Output: Quantum geometry notation sheet.
- Contact action: No emails.

Day 37 - QGT computation

- Work block: Compute QGT numerically for your QAOA ansatz at selected parameters.
- Output: Matrix output and conditioning diagnostic.
- Contact action: No emails.

Day 38 - Natural-gradient optional test

- Work block: If QGT works, compare ordinary parameter update with natural-gradient-style update. If not, just report QGT diagnostics.
- Output: Optional plot: ordinary vs geometry-aware update.
- Contact action: No emails.

Day 39 - D-module minimal theory

- Work block: Read only Weyl algebra, left ideals, and differential operator examples. Do not read derived categories.
- Output: D-module notation sheet.
- Contact action: No emails.

Day 40 - Linearized equation

- Work block: Derive a simplified scalar linear ODE from radial perturbation/geodesic deviation around a reference orbit. Clear denominators.
- Output: ODE written as $L \xi = 0$ with polynomial coefficients if possible.
- Contact action: Draft D-module email.

Day 41 - Macaulay2/Sage computation

- Work block: Install or use available tool. Define Weyl algebra and left ideal generated by L . Compute whatever is realistically available.
- Output: symbolic/dmodule_jacobi.m2 and output log.
- Contact action: Send 1-2 D-module/algebraic-analysis emails only if L is written clearly.

Day 42 - Checkpoint

- Work block: Write what the D-module appendix actually proves/does not prove.
- Output: Week 6 checkpoint.
- Contact action: No follow-up yet.

Week 7 - Experiments, ablations, and report body

Goal: End the week with all core experiments complete and most of the main report drafted.

Day 43 - Experiment matrix

- Work block: Define 6-10 experiments: time step, orbit type, noise level, burn budget, horizon length, QAOA depth.
- Output: Experiment plan table.
- Contact action: No emails.

Day 44 - Run simulation sweeps

- Work block: Run geodesic/autonomy experiments. Save all parameters.
- Output: results/ folder with CSVs.
- Contact action: No emails.

Day 45 - Run quantum sweeps

- Work block: Run QAOA/brute force/greedy comparisons on selected instances.
- Output: Benchmark table.
- Contact action: No emails.

Day 46 - Figures day

- Work block: Create final figures: orbit, drift, control improvement, QUBO/QAOA comparison, QGT matrix/condition.
- Output: At least 5 publication-quality figures.
- Contact action: No emails.

Day 47 - Report writing block

- Work block: Write introduction, methods, simulator, autonomy, quantum module sections.
- Output: Main report 12-18 pages draft.
- Contact action: Send short update to any helper who already replied.

Day 48 - External feedback packet

- Work block: Prepare 1-page summary + 2 figures + precise questions.
- Output: Feedback packet v1.
- Contact action: Send to responders; optionally send one carefully chosen new email if a category is missing.

Day 49 - Checkpoint and triage

- Work block: Decide what to cut. Freeze experiments. No new technical modules after today unless required for correctness.
- Output: Week 7 checkpoint.
- Contact action: No follow-up yet.

Week 8 - Final packaging, preprint note, and supervisor outreach

Goal: End the week with a polished GitHub repo, main report, short note, and supervisor-facing email package.

Day 50 - Freeze code

- Work block: Clean repository, remove dead scripts, write build instructions, add sample commands.
- Output: README v1; reproducibility commands.
- Contact action: No emails.

Day 51 - Main report polish

- Work block: Complete 20-30 page report. Add limitations and no-overclaiming section.
- Output: Main report near-final.

- Contact action: No emails.

Day 52 - Preprint-style note

- Work block: Compress project into 6-8 pages: problem, method, results, limits, next steps.
- Output: Short note draft.
- Contact action: No emails.

Day 53 - Supervisor package

- Work block: Create one-page research summary, one-page reading record, one-page open questions list.
- Output: Supervisor package PDF.
- Contact action: Send to 3-4 targeted senior contacts; ask for feedback/referral.

Day 54 - Fix based on replies

- Work block: Incorporate easy corrections. Improve README and report abstract.
- Output: Final technical artifacts.
- Contact action: Respond to any replies within 24-48 hours.

Day 55 - Final archive

- Work block: Tag GitHub release v0.1. Export report PDF and note PDF. Check that figures reproduce.
- Output: Release checklist complete.
- Contact action: No new cold emails.

Day 56 - Reflection and next-stage plan

- Work block: Write final reflection: what you learned, what failed, what becomes a Master project.
- Output: Final reflection + next semester plan.
- Contact action: Schedule follow-up meetings; send thank-you notes to helpers.

7. Weekly Deliverables and Checkpoints

Week	Deliverable	Checkpoint question
Week 1	Repo skeleton, RK integrator, first Schwarzschild RHS, GR notation sheet	Do I know exactly what equations I am integrating?
Week 2	Validated geodesic solver, Newtonian comparison, first orbit figures, first GR outreach	Can I explain what is relativistic in the simulation?
Week 3	Toy autonomy loop, noise/perturbation model, greedy controller	Does the controller actually improve a measurable target?
Week 4	QUBO formulation, brute force solver, first QI outreach	Does my binary optimization problem match the burn-planning problem?
Week 5	QAOA simulation, baseline table, honest quantum section draft	Can I compare QAOA without claiming speedup?
Week 6	QGT computation, D-module notation sheet, Weyl algebra toy computation	Is the D-module appendix small, linear, and honest?
Week 7	Experiment sweeps, figures, main report draft	Are the results reproducible and plotted clearly?
Week 8	Final repo, report, note, supervisor package	Could I send this without embarrassment?

Weekly checkpoint template

1. What definitions can I state from memory?
2. What derivation can I reproduce without notes?
3. What code compiles and produces a saved result?
4. What figure/table did I produce this week?
5. What is still vague?
6. What is the one sentence connecting this week to the final paper?
7. What must be cut next week to stay on schedule?

8. Code Repository Blueprint

```
relativistic-quantum-autonomy/  
  README.md  
  CMakeLists.txt  
  cpp/  
    include/  
      spacetime/Schwarzschild.hpp  
      integrators/RK4.hpp  
      autonomy/Planner.hpp  
      optimization/QUBO.hpp  
      quantum/QAOA.hpp  
    src/  
      spacetime/Schwarzschild.cpp  
      integrators/RK4.cpp  
      autonomy/Planner.cpp  
      optimization/QUBO.cpp  
      quantum/QAOA.cpp  
    tests/  
      test_integrator.cpp  
      test_qubo_energy.cpp  
      test_conserved_quantities.cpp  
  experiments/  
    01_geodesic_validation/  
    02_newtonian_vs_schwarzschild/  
    03_autonomy_loop/  
    04_qubo_qaoa/
```

```

05_quantum_geometry/
symbolic/
  dmodule_jacobi.m2
  sage_schwarzschild_check.sage
scripts/
  plot_orbit.py
  plot_benchmarks.py
report/
  main.tex or main.md
  figures/
notes/
  weekly_checkpoints/

```

Minimum tests

- Integrator test: harmonic oscillator stays bounded and converges when dt decreases.
- Conservation test: energy/angular momentum drift decreases with dt.
- QUBO test: Hamiltonian energy equals QUBO cost for all bitstrings on a tiny instance.
- Planner test: controlled trajectory beats uncontrolled trajectory on at least one target metric.
- Reproducibility test: one command regenerates at least the main experiment table.

9. Report and Preprint Blueprint

20-30 page report structure

Section	Content
Abstract	State the toy problem, methods, and no-overclaiming limits.
1. Introduction	Why relativistic navigation/autonomy and quantum optimization are a meaningful toy interface.
2. Mathematical background	Schwarzschild metric, geodesics, conserved quantities, QUBO, QAOA, quantum geometric tensor, D-module mini-background.
3. Simulator	C++ architecture, numerical method, validation tests.
4. Autonomy loop	State, observation/noise model, correction actions, cost function.
5. Quantum optimization	QUBO construction, brute force, greedy baseline, QAOA implementation.
6. Quantum geometry	QGT computation and interpretation.
7. D-module appendix	Linearized equation, Weyl algebra object, computation log.
8. Experiments	Figures and tables.
9. Limitations	No real mission, no speedup, no full control theory, no full D-module theory.
10. Outlook	Master project continuation ideas.

6-8 page preprint-style note

- Page 1: title, abstract, motivation, exact scope.
- Page 2: mathematical model and simulator.
- Page 3: autonomy and QUBO formulation.
- Page 4: QAOA benchmark.
- Page 5: quantum geometry and/or D-module appendix.
- Page 6: results and limitations.
- Pages 7-8: references and future work if needed.

10. Email Templates

Template A - GR / numerical relativity sanity check

Subject: Question about a small Schwarzschild geodesic simulation project

Dear Dr./Professor [Name],

I hope you are well. I am a mathematics student preparing for a Master's direction in quantum information and mathematical physics. This summer I am building a small computational project: a C++ toy simulator for a spacecraft/test particle in Schwarzschild spacetime, with a simple autonomous correction-burn loop.

The project is intentionally modest. I am not trying to build a real spacecraft guidance system. My current goal is to validate the geodesic propagation, compare it with a Newtonian baseline, and then use the resulting toy model as input for a small quantum-optimization module.

Would you be open to a short sanity-check question, or could you suggest a PhD student/postdoc who might be willing to tell me whether my Schwarzschild/numerical framing is reasonable? I can send a one-page summary and one validation figure.

Best regards,
Robin

Template B - Quantum information / QAOA feedback

Subject: Feedback request on a toy QAOA formulation for relativistic burn planning

Dear [Name],

I am working on a small summer project at the interface of quantum information and mathematical physics. The current prototype simulates a toy spacecraft in Schwarzschild spacetime, then formulates a short-horizon correction-burn planning problem as a small QUBO. I solve the QUBO by brute force and a greedy baseline, then test a simulated QAOA circuit.

I am not claiming quantum advantage. My goal is to learn hands-on quantum optimization and produce a mathematically honest report. Would you be willing to give brief feedback on whether the QUBO/QAOA framing is sensible, or suggest someone in the Paris quantum-information ecosystem who might be a good fit for a 15-minute sanity check?

Best regards,
Robin

Template C - D-module / Weyl algebra question

Subject: Small Weyl-algebra question from a linearized geodesic-deviation toy model

Dear Dr./Professor [Name],

I am a mathematics student working on a small summer computational project. The main project is a toy relativistic-autonomy simulator, but I am trying to add a very small algebraic-analysis appendix. I linearize a simplified geodesic-deviation/radial perturbation equation and write it as a differential operator L acting on a perturbation ξ .

My intended D-module component is deliberately modest: define the Weyl algebra, form the left ideal generated by L , and compute a small symbolic example in Macaulay2/Sage. I am not trying to use deep microlocal theory.

Would you be willing to tell me whether this is a mathematically honest way to include a computational D-module example, or suggest a PhD student/postdoc who might be willing to comment?

Best regards,
Robin

Template D - Follow-up

Subject: Follow-up: [same subject]

Dear [Name],

I hope you are well. I wanted to follow up once on the short message below. I completely understand if you are busy. Even a pointer to a relevant student, postdoc, note, or warning about the project framing would be very helpful.

Thank you again for your time.

Best regards,
Robin

Template E - Thank-you after receiving help

Subject: Thank you

Dear [Name],

Thank you very much for your reply and suggestions. I will adjust the project scope accordingly, especially [specific point they made]. I appreciate the warning about [specific issue], and I will make sure not to overclaim it in the report.

If I manage to produce a clean final note, may I send you the short version at the end of the summer?

Best regards,
Robin

11. Failure Modes and Cut Rules

Failure modes

- Trying to learn all of GR before coding. Cut: learn only Schwarzschild geodesics and diagnostics.
- Spending too long installing Qulacs/CUDA-Q. Cut: write a tiny state-vector simulator for ≤ 6 qubits.
- Letting D-modules become pure abstraction. Cut: one linear ODE, one Weyl algebra, one computation.
- Trying to do all of control theory. Cut: use a greedy short-horizon planner.
- Writing no report until Week 8. Cut: write every week, even rough notes.
- Overclaiming quantum advantage. Cut: say benchmark/prototype only.
- Emailing too early with no artifact. Cut: wait until you have one page and one plot.

If overloaded

Situation	Cut rule
Week 1-2 overloaded	Keep geodesic simulator and diagnostics; cut proper-time details.
Week 3 overloaded	Keep greedy controller; cut estimator/Kalman filtering.
Week 4-5 overloaded	Keep QUBO + brute force; implement only $p=1$ QAOA.
Week 6 overloaded	Keep either QGT or D-module; do not force both if the core report is weak.
Week 7 overloaded	Freeze experiments; prioritize figures and report clarity.
Week 8 overloaded	Deliver GitHub + 15-page report + one-page summary; cut preprint polish.

12. Source and Contact Reference List

Internal project sources from your uploaded files:

- The_Map.pdf - source for the discipline of weekly writing, prove/quote/compute split, and supervisor package mentality.
- advanced_numerical_analysis_problems.pdf - source for the numerical-analysis bridge from finite differences to operator methods and Riemannian/geodesic numerical error.
- coursebook.pdf and Proj.pdf - source for weighted manifolds, weighted divergence/Laplacian, Gaussian/OU diffusion, and quantum-geometric examples.

Public sources and contact pages checked while preparing this document:

- LIP6 Quantum Information Team: <https://qi.lip6.fr/>
- Damian Markham personal page and current group listing: <https://damianmarkham.weebly.com/>
- Marc-Olivier Renou page / PhiQuS information: <https://marcolivierrenou.com/>
- PhiQuS / LIX page: <https://www.lix.polytechnique.fr/phiqus/>
- SYRTE/LTE Observatoire de Paris: <https://syrtel.obspm.fr/spip/?lang=en>
- ONERA DTIS research units / IGNC: <https://www.onera.fr/en/dtis-research-units>
- Ericourgoulhon homepage / SageManifolds and numerical relativity resources: <https://people-lux.obspm.fr/ericourgoulhon/en/>
- Pierre Schapira homepage: <https://webusers.imj-prg.fr/~pierre.schapira/>
- Pierre Schapira D-module notes: <https://webusers.imj-prg.fr/~pierre.schapira/LectNotes/Dmod.pdf>
- Claude Sabbah page: <https://perso.pages.math.cnrs.fr/users/claude.sabbah/>
- Qulacs documentation: <https://docs.qulacs.org/en/latest/index.html>
- QAOA original paper: <https://arxiv.org/abs/1411.4028>
- NASA smallsat GNC overview: <https://www.nasa.gov/smallsat-institute/sst-soa/guidance-navigation-and-control/>

Final reminder

This is not a plan to become an expert in GR, quantum algorithms, D-modules, and autonomy in eight weeks. It is a plan to build one coherent artifact that proves you can learn, implement, simulate, write, and ask mathematically honest questions.